The Smart Predictive Geofencing System for Animals is an hardware-based animal monitoring and boundary-alert system designed to improve the safety and tracking of livestock and domestic animals. The project integrates GPS-based location tracking, motion sensing, and intelligent risk assessment into a compact, wearable collar enclosure.

The system uses a NEO-6M GPS module to obtain the animal's geographical coordinates, speed, and direction, while an MPU6050 Inertial Measurement Unit (IMU) detects movement, acceleration, and rotational motion. An ESP32 microcontroller serves as the central processing unit, collecting sensor data, evaluating movement patterns, calculating the animal's distance from a predefined geofence boundary, and determining its potential escape risk.

Based on the processed information, the system classifies the animal's status into four levels: Safe, Caution, High Risk, and Escaped/Critical. Three LED indicators provide visual alerts, while a buzzer generates audible warnings when the animal approaches or crosses the designated boundary. The proposed architecture also supports optional Wi-Fi or Bluetooth connectivity for displaying location, risk level, and boundary status on a smartphone or computer dashboard.

The hardware is designed around a rechargeable lithium-ion battery, a TP4056 charging module, and a protective enclosure attached to an adjustable collar strap. Its compact construction aims to provide portability, durability, and suitability for extended animal monitoring.

The project's principal objective is to develop a smart, predictive, and wearable geofencing solution that combines location awareness and movement analysis to facilitate early warnings, reduce animal escape incidents, and support efficient livestock and pet management.